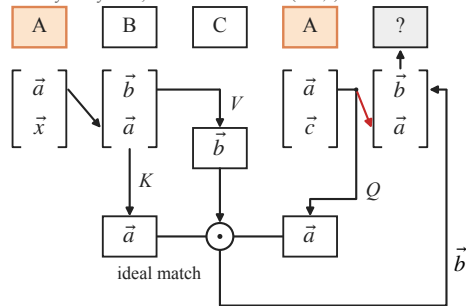


(a) look-back pairing

pair at B: key from A, value from B

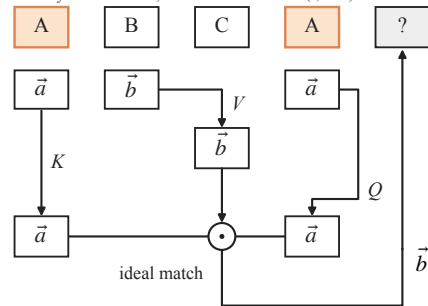
built by a key shift, or a conv on x over $(t-1, t)$



(b) look-ahead pairing

pair at A: key from A, value from B

built by a value shift, or a conv on x over $(t, t+1)$



(c) Matched temporal windows

same backbone; 144k allocated parameters

Memory input (conv on x)

single frame (t)

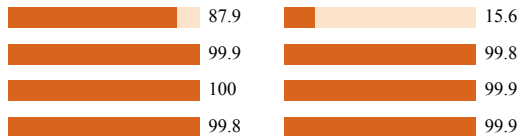
$(t-1, t)$: look-back

$(t, t+1)$: look-ahead

$(t-1, t, t+1)$: model learns look-ahead

learned PE

no learned PE



L3 strict SeqAcc (%); three-seed means. Both directions work (gap < 0.2 pp); given both, attention picks look-ahead.